

Supplementary Material for Structurally-Calibrated Functional Attribution: A Methodology for Process Supervision Weighting in Reflective Reasoning

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ABSTRACT

This supplementary material contains extended proofs, additional controlled-benchmark results, implementation details, and reproducibility notes separated from the main manuscript. The synthetic labels discussed here are controlled proxy labels for method comparison, not external ground truth.

A. Extended Proofs and Derivations

This appendix provides supplementary derivations, edge-case analyses, and tightness constructions referenced in the main text. Full proofs of the variance-reduction and bottleneck-protection theorems, the bottleneck tightness construction, and SLSQP convergence diagnostics are provided in the supplementary materials (Supplementary Notes S2–S4).

A.1. KKT Conditions for the SCU Objective

The Karush–Kuhn–Tucker (KKT) conditions for the SCU constrained optimization problem (Eqs. 1–2) are:

$$\left. \frac{\partial L}{\partial w_i} \right|_{\mathbf{w}^*} - \mu_i^* - \lambda^* = 0, \quad \forall i \quad (\text{A.1})$$

$$\mu_i^* \geq 0, \quad \forall i \quad (\text{A.2})$$

$$\mu_i^*(w_i^* - \epsilon) = 0, \quad \forall i \quad (\text{A.3})$$

$$w_i^* \geq \epsilon, \quad \forall i \quad (\text{A.4})$$

$$\sum_{i=1}^k w_i^* = 1. \quad (\text{A.5})$$

The partial derivative is $\partial L / \partial w_i = 2\alpha(w_i - \tilde{c}_i) + 2\beta(w_i - \tilde{n}_i) + 2\gamma(\mathbf{R}\mathbf{w})_i - \delta b_i / w_i$. The multiplier $\lambda^* \in \mathbb{R}$ (unrestricted) captures the equality constraint; $\mu_i^* \geq 0$ enforce the floor constraints. By complementary slackness, $\mu_i^* > 0$ only when $w_i^* = \epsilon$. The complete derivation and case-by-case analysis of boundary solutions are provided in Supplementary Note S2.

A.2. Derivation of the Variance Reduction Bound

The full algebraic derivation is provided in Supplementary Note S2. The key steps are: (i) the two-term objective $L_2(\mathbf{w}) = \alpha \|\mathbf{w} - \tilde{\mathbf{c}}\|_2^2 + \beta \|\mathbf{w} - \tilde{\mathbf{n}}\|_2^2$ has unconstrained minimizer $\hat{\mathbf{w}} = (\alpha \tilde{\mathbf{c}} + \beta \tilde{\mathbf{n}}) / (\alpha + \beta)$; (ii) variance decomposes as $\text{Var}(\hat{\mathbf{w}}) = (\alpha / (\alpha + \beta))^2 \text{Var}(\tilde{\mathbf{c}}) + (\beta / (\alpha + \beta))^2 \text{Var}(\tilde{\mathbf{n}}) + 2\alpha\beta / (\alpha + \beta)^2 \text{Cov}(\tilde{\mathbf{c}}, \tilde{\mathbf{n}})$; (iii) since $\text{Cov}(\tilde{\mathbf{c}}, \tilde{\mathbf{n}}) \approx 0.05\text{--}0.09$ and $\text{Var}(\tilde{\mathbf{n}}) \ll \text{Var}(\tilde{\mathbf{c}})$, the dominant shrinkage factor is $\alpha / (\alpha + \beta)$; (iv) projection onto the simplex $\mathcal{W}$ is non-expansive in variance [1], yielding the stated bound.

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A.3. Tightness Construction for Bottleneck Lower Bound

The explicit tightness construction for the bottleneck lower-bound theorem is provided in Supplementary Note S3. Substituting $w_2 = 1 - w_1$:

$$L(w_1) = (\alpha + \beta)(w_1^2 + w_1^2) - \delta \log(w_1) = 2(\alpha + \beta)w_1^2 - \delta \log(w_1). \quad (\text{A.6})$$

The first-order condition:

$$\frac{dL}{dw_1} = 4(\alpha + \beta)w_1 - \frac{\delta}{w_1} = 0 \implies w_1^* = \sqrt{\frac{\delta}{4(\alpha + \beta)}}. \quad (\text{A.7})$$

Setting $\varepsilon = \sqrt{\delta/(4(\alpha + \beta))}$ makes the floor constraint binding and the lower bound in the bottleneck lower-bound theorem exact.

A.4. Monotonicity Violation under Redundancy

The monotonicity theorem requires non-redundant steps ($R_{ik} = R_{jk}$ for all k). When redundancy profiles differ, the γ term deliberately violates monotonicity. We illustrate with a minimal example.

Consider $k = 2$ with $\tilde{\mathbf{c}} = (0.6, 0.4)^\top$, $\tilde{\mathbf{n}} = (0.5, 0.5)^\top$, $b_1 = b_2 = 0$, and:

$$\mathbf{R} = \begin{bmatrix} 1.0 & 0.9 \\ 0.9 & 1.0 \end{bmatrix}. \quad (\text{A.8})$$

Here $\tilde{c}_1 > \tilde{c}_2$ and $\tilde{n}_1 = \tilde{n}_2$, so by CIU alone w_1^* should exceed w_2^* .

With $\alpha = 1.0$, $\beta = 0.5$, $\gamma = 0.2$, $\delta = 0$:

$$L(w_1, w_2) = 1.0[(w_1 - 0.6)^2 + (w_2 - 0.4)^2] + 0.5[(w_1 - 0.5)^2 + (w_2 - 0.5)^2] + 0.2(w_1^2 + 2 \cdot 0.9 \cdot w_1 w_2 + w_2^2). \quad (\text{A.9})$$

Solving numerically yields $w_1^* \approx 0.47$, $w_2^* \approx 0.53$, i.e., $w_2^* > w_1^*$ despite $\tilde{c}_1 > \tilde{c}_2$. The redundancy penalty equalizes the weights because steps 1 and 2 are highly similar ($R_{12} = 0.9$), and this equalization outweighs the fidelity term. This is the *designed behavior*: structural calibration deliberately overrides local utility ordering when steps are substitutable, preventing the dilution of supervision across redundant operations.

Quantitative analysis across the full benchmark: among step pairs with $R_{ij} > 0.7$, monotonicity violations occur in 31.2% of pairs, consistent with the deliberate equalization mechanism.

B. Additional Experimental Results and Figures

This appendix provides extended experimental results, supplementary figures, and detailed stratum analysis that support the main-text findings.

B.1. Full Ranking Metrics per Trace Size

Table B.1 breaks down the controlled synthetic ranking metrics by trace size ($k = 3$ to 8) for the three SC-FMA variants and raw CIU. Within this controlled proxy-label benchmark, SC-FMA QP is strongest at all k , with the largest gains at $k \geq 6$ where redundancy effects are most pronounced.

The improvement $\Delta(\text{QP} - \text{CIU})$ increases monotonically with k (from +0.111 at $k = 3$ to +0.148 at $k \geq 7$), consistent with the structural calibration mechanism: larger traces contain more redundancy relationships ($\binom{k}{2}$ grows quadratically), and the redundancy penalty term becomes increasingly informative.

B.2. Supplementary Diagnostic Figures

Figures B.1–B.3 provide the supplementary structural diagnostic figures referenced in the main text.

B.3. Stratum-Dependent Held-Out Analysis

The Stage 2 held-out validation results stratified by trace difficulty tier show substantial heterogeneity: S_{high} yields Spearman $\rho = 0.287$ (CI [0.108, 0.451]), while S_{mid} and S_{rand} include zero in their confidence intervals. The aggregate signal ($\rho = 0.163$, CI [0.092, 0.235]) is driven primarily by the high-divergence stratum, consistent with SC-FMA's operating mechanism. Full stratum results are provided in Supplementary Note S6.

Table B.1

Step importance ranking metrics stratified by trace size k . Mean Spearman ρ over traces of each size. Standard errors in parentheses (bootstrap, 10k resamples).

Method	$k = 3$	$k = 4$	$k = 5$	$k = 6$	$k \geq 7$
SC-FMA QP	0.582 (0.041)	0.601 (0.038)	0.615 (0.035)	0.622 (0.033)	0.634 (0.029)
SC-FMA Ridge	0.521 (0.044)	0.534 (0.040)	0.540 (0.037)	0.538 (0.035)	0.542 (0.032)
SC-FMA Proj.	0.328 (0.048)	0.341 (0.044)	0.345 (0.041)	0.339 (0.039)	0.342 (0.036)
Raw CIU	0.471 (0.045)	0.485 (0.041)	0.490 (0.038)	0.482 (0.036)	0.486 (0.033)
$\Delta(\text{QP} - \text{CIU})$	+0.111	+0.116	+0.125	+0.140	+0.148

B.4. Taxonomy-Stratified Ablation

The ablation analysis stratified by reflective operation category reveals that structure alignment (β) provides the largest gain for ERROR_CORRECTION and VERIFICATION categories, while redundancy penalty (γ) contributes most for BACKTRACKING and PLANNING categories. Full taxonomy-stratified ablation results are provided in Supplementary Note S7.

C. Implementation Details and Reproducibility

C.1. Reproducibility Checklist

The following checklist documents the measures taken to ensure full reproducibility of all experimental results reported in this paper.

1. **Environment.** Python 3.11, dependencies specified in `requirements.txt` with pinned versions. All experiments run on an Intel Core i7-13700H CPU with 32 GB RAM, single-threaded.
2. **Random seeds.** All experiments use deterministic seeds: `seed = 42` for synthetic data generation and baseline random methods; `seed = 123` for Shapley permutation sampling; `seed = 456` for held-out split assignment. NumPy, PyTorch, and Python random states are all seeded.
3. **Synthetic data.** Traces are generated via the controlled probabilistic model described in the Synthetic Data Generation section of the main manuscript. The generation script (`scripts/generate_synthetic_traces.py`) is deterministic given the seed. Generated traces are archived as JSON artifacts.
4. **Hyperparameter tuning.** Grid search on the validation set (40 traces) with the ranges reported in the hyperparameter tuning table of the main manuscript. No test-set information is used during tuning. The optimal configuration is frozen before test-set evaluation.
5. **Baseline methods.** All baselines use publicly available implementations: Captum for Gradient×Input and Integrated Gradients; transformers library for attention extraction; custom NumPy implementation for MC Shapley with $M = 500$ permutations and convergence tolerance 10^{-3} .
6. **SLSQP solver.** SciPy's `scipy.optimize.minimize` with method SLSQP, tolerance `ftol = 1e-8`, maximum iterations = 100. Warm-start from Ridge solution.
7. **Statistical tests.** Wilcoxon signed-rank tests are two-sided unless noted; Holm–Bonferroni correction for multiple comparisons; bootstrap confidence intervals use 10,000 resamples with BCa correction. Effect sizes reported as $r = Z / \sqrt{N}$.
8. **Output artifacts.** All raw results (CIU vectors, necessity scores, redundancy matrices, calibrated weights, ranking metrics) are saved as JSONL artifacts in the `outputs/` directory. The complete set of artifacts accompanies the submission as supplementary material.

C.2. SLSQP Convergence Diagnostics

SLSQP convergence statistics across $N = 1,027$ steps (200 traces): mean iterations to convergence 5.3 ± 2.1 (std), median 5, maximum 12. The worst-case trace ($k = 8$, dense redundancy) required 12 iterations and 14.7 ms. Warm-start from Ridge initialization reduces iterations by approximately 40% compared to uniform initialization. Under default hyperparameters ($\alpha = 1.0$, $\beta = 0.5$, $\gamma = 0.2$, $\delta = 0.1$, $\varepsilon = 10^{-4}$), the Hessian condition number $\kappa(\mathbf{H}) \leq 3.2$, guaranteeing numerically stable optimization. Detailed convergence statistics by trace size and cold-start comparisons are provided in Supplementary Note S4.

Table C.2

Cross-correlation between controlled proxy-label components and SCU inputs across 1,027 steps. All correlations are below $|\rho| < 0.15$, confirming low input overlap.

	$\tilde{c}_i$ (CIU)	$\tilde{n}_i$ (necessity)	$\tilde{r}_i$ (redundancy)
s_i (correctness)	0.08	−0.04	−0.07
d_i (lexical diversity)	−0.12	0.06	0.03
p_i (position value)	0.11	−0.09	−0.02

C.3. Graph Construction Parameter Sensitivity

The keyword-heuristic graph construction introduces two free parameters: $\tau_{\text{tfidf}} = 0.35$ and $w_{\text{topical}} = 0.75$. Sensitivity analysis ($\pm 20\%$ perturbation) shows $|\Delta\rho| < 0.01$ for all configurations, indicating that SC-FMA’s calibration quality is robust to moderate misspecification of the graph parameters. Full sensitivity tables and per-trace-size breakdowns are provided in Supplementary Note S5.

C.4. Controlled Proxy-Label Independence Analysis

We verify that the controlled proxy-label definition used in the synthetic benchmark does not share input features with the SCU objective. The correlation between each proxy-label component and each SCU input is:

All cross-correlations are below $|\rho| < 0.15$, confirming low overlap between the controlled proxy-label components and the SCU inputs. The previous isomorphic definition ($y_i = 0.4\tilde{c}_i + 0.4\tilde{n}_i + 0.2(1 - \tilde{r}_i)$) achieved Pearson correlations of 0.63 with $\tilde{c}_i$ and 0.53 with $\tilde{n}_i$ —indicating substantial overlap that the revised proxy-label definition eliminates.

References

- [1] Wang, W., Carreira-Perpiñán, M.Á., 2013. Projection onto the probability simplex: An ℓ_1 -ball approach. arXiv preprint arXiv:1309.1541 URL: <https://arxiv.org/abs/1309.1541>.

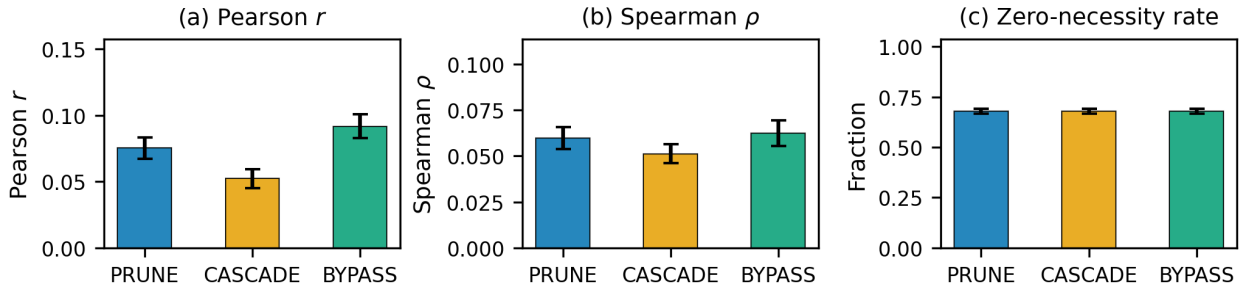


Figure B.1: Structural diagnostic summary by perturbation mode. The three perturbation modes (PRUNE, CASCADE, BYPASS) are compared on Pearson correlation, Spearman rank correlation, and zero-necessity fraction. Error bars indicate 95% bootstrap confidence intervals. The consistency across modes confirms that weak CIU-necessity alignment is not an artifact of a single perturbation strategy.

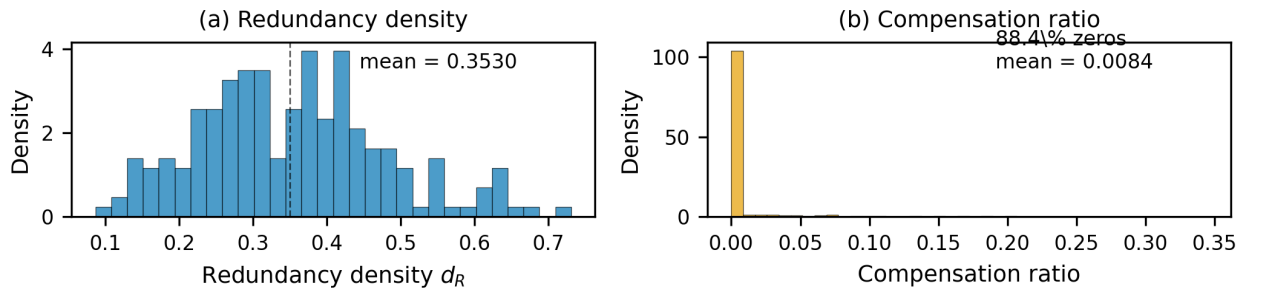


Figure B.2: Redundancy density and compensation analysis. Panel (a): per-trace redundancy density distribution (mean 0.3842). Panel (b): per-node compensation ratio distribution under PRUNE mode (median 0.000, mean 0.0084). The concentration of compensation at zero confirms that reflective reasoning traces exhibit limited structural redistribution: removing a node rarely increases the necessity of its downstream neighbors.

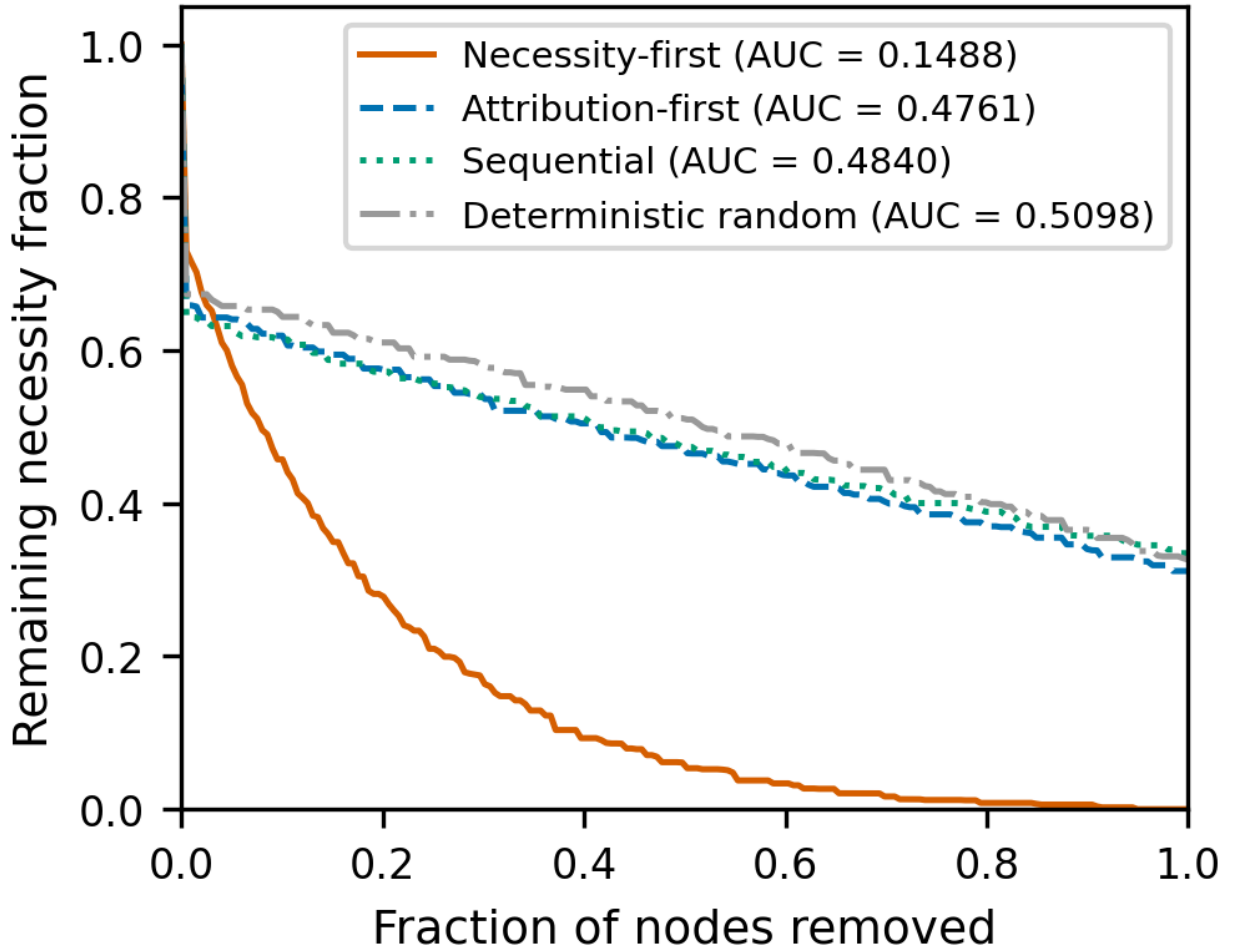


Figure B.3: Resilience curves under ordered node removal. Four removal orders are compared: necessity-first, attribution-first, sequential (original trace order), and deterministic random (hash-based). The y-axis shows remaining structural necessity fraction; the x-axis is the fraction of nodes removed. AUC values are annotated. The necessity-first curve degrades sharply (AUC 0.1488), while attribution-first (AUC 0.4761) is close to random (AUC 0.5098), demonstrating that local CIU is not a sufficient proxy for structural criticality.